# Daily Algebra Drill #6 — Answers & Investigations

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| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.*

## Section A Rapid Recognition

**1.** Factorise: $x^3 + 5x^2 + 6x$.

**Answer:** $x(x+2)(x+3)$

**Method:** Factor out $x$: $x(x^2+5x+6) = x(x+2)(x+3)$. Always extract common factors before any other step.

> **Investigate further:** The roots are $0, -2, -3$. The product $0 \times (-2) \times (-3) = 0$ confirms $c = 0$ in Vieta. Now: for $x^3 + ax^2 + bx = 0$ (no constant term), $x = 0$ is always a root. What does this mean about the graph of $y = x^3 + 5x^2 + 6x$?

**2.** Given $x + \dfrac{1}{x} = \sqrt{5}$, find $x^2 + \dfrac{1}{x^2}$.

**Answer:** 3

**Method:** $\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2} = 5$, so $x^2 + \frac{1}{x^2} = 3$.

> **Investigate further:** Note $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = 5 - 2 = 3 < 5$. Continue the chain to find $x^3 + \frac{1}{x^3} = (\sqrt{5})(3) - \sqrt{5} = 2\sqrt{5}$ and $x^4 + \frac{1}{x^4} = 3^2 - 2 = 7$.

**3.** State the coefficient of $xy^2$ in $(x+y)^3$.

**Answer:** 3

**Method:** $\binom{3}{1} = 3$ (choosing 1 factor to contribute $x$, the rest give $y$).

> **Investigate further:** The binomial theorem: $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$. The coefficient of $x^k y^{n-k}$ is $\binom{n}{k}$. Use this to find the coefficient of $x^2y^3$ in $(2x-y)^5$.

**4.** Evaluate $\dfrac{1}{\sqrt{2}-1} - \dfrac{1}{\sqrt{2}+1}$.

**Answer:** 2

**Method:** Rationalise both fractions: $\frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} - \frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)} = (\sqrt{2}+1) - (\sqrt{2}-1) = 2$.

> **Investigate further:** Rationalising: $\frac{1}{\sqrt{n}-1} = \frac{\sqrt{n}+1}{n-1}$ and $\frac{1}{\sqrt{n}+1} = \frac{\sqrt{n}-1}{n-1}$. Their difference is $\frac{2}{n-1}$. Verify at $n = 2$: $\frac{2}{1} = 2$. ✓Now evaluate $\frac{1}{\sqrt{5}-2} - \frac{1}{\sqrt{5}+2}$.

**5.** If $x = 2$ is a repeated root of $f(x) = x^3 + ax + b$, find $a$ and $b$.

**Answer:** $a = -12,\ b = 16$

**Method:** $(x-2)^2$ divides $f(x)$, so $f(x) = (x-2)^2(x-r)$ for some $r$. Expanding: $x^3 - (r+4)x^2 + (4r+4)x - 4r$. Matching $f(x) = x^3 + ax + b$ (no $x^2$ term): $r + 4 = 0 \Rightarrow r = -4$. Then $a = 4(-4) + 4 = -12$ and $b = -4(-4) = 16$.

> **Investigate further:** Alternatively: if $x = 2$ is a repeated root, then $f(2) = 0$ AND $f'(2) = 0$. $f'(x) = 3x^2 + a$. $f'(2) = 12 + a = 0 \Rightarrow a = -12$. Then $f(2) = 8 - 24 + b = 0 \Rightarrow b = 16$. This derivative approach is the standard method when calculus is allowed.

**6.** Simplify: $\left(\dfrac{2x}{y}\right)^3 \div \dfrac{4x^2}{y^2}$.

**Answer:** $\dfrac{2x}{y}$

**Method:** $\frac{8x^3}{y^3} \div \frac{4x^2}{y^2} = \frac{8x^3}{y^3} \cdot \frac{y^2}{4x^2} = \frac{2x}{y}$.

> **Investigate further:** This simplification reveals $\left(\frac{2x}{y}\right)^3 \div \left(\frac{2x}{y}\right)^2 = \frac{2x}{y}$. The pattern: $t^3 \div t^2 = t$ for any $t = \frac{2x}{y}$. Recognising the hidden variable $t$ saves the expansion step entirely.

**7.** Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.

**Answer:** 30

**Method:** $\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta) = 6 \times 5 = 30$.

> **Investigate further:** Factor before substituting. The expression $\alpha^2\beta + \alpha\beta^2$ has an obvious common factor $\alpha\beta$ that you should see immediately. Now find $\alpha^3\beta^2 + \alpha^2\beta^3 = \alpha^2\beta^2(\alpha + \beta)$.

**8.** Factorise: $4a^2 - 4ab + b^2$.

**Answer:** $(2a - b)^2$

**Method:** Perfect square trinomial with $(\alpha - \beta)^2$ form: $\alpha = 2a$, $\beta = b$.

> **Investigate further:** Perfect square recognition: $px^2 + qxy + ry^2$ is a perfect square iff $q^2 = 4pr$. Check: $(-4)^2 = 16 = 4 \times 4 \times 1$. ✓Identify all perfect square trinomials of the form $4a^2 + kab + 9b^2$.

**9.** Simplify $2^{10} - 1$ as a product of three factors greater than 1.

**Answer:** $3 \times 11 \times 31$

**Method:** $2^{10} - 1 = 1023 = (2^5 - 1)(2^5 + 1) = 31 \times 33 = 31 \times 3 \times 11$.

> **Investigate further:** $2^{10} - 1 = (2^2 - 1)(2^8 + 2^6 + 2^4 + 2^2 + 1) = 3 \times 341 = 3 \times 11 \times 31$. Alternatively use the factorisation chain $2^{10} - 1 = (2^5 - 1)(2^5 + 1)$. Which approach is faster? Use Mersenne numbers $M_p = 2^p - 1$: are they always prime for prime $p$? (No: $2^{11} - 1 = 2047 = 23 \times 89$.)

**10.** Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

**Answer:** 0

**Method:** $f(0) = 0^2 - 3(0) + 2 = 2$. Then $f(f(0)) = f(2) = 2^2 - 3(2) + 2 = 0$.

> **Investigate further:** $f(f(0)) = 0$ means 0 is a period-2 point of $f$: $f(0) = 2$ and $f(2) = 0$, so $f(f(0)) = 0$. Explore: is 0 a fixed point? ($f(0) = 2 \neq 0$, so no.) Find all fixed points of $f$ (where $f(x) = x$: $x^2 - 4x + 2 = 0$, $x = 2 \pm \sqrt{2}$). What does the graph of $y = f(f(x))$ look like?

## Section B Manipulation Drills

**1.** Find $A, B, C$ such that $x^2 + 3x + 5 \equiv A(x + 1)^2 + B(x + 1) + C$.

**Answer:** $A = 1,\ B = 1,\ C = 3$

**Method:** Expand the right: $A(x^2 + 2x + 1) + B(x + 1) + C = Ax^2 + (2A + B)x + (A + B + C)$. Compare: $A = 1$, $2A + B = 3 \Rightarrow B = 1$, $A + B + C = 5 \Rightarrow C = 3$. Alternatively substitute $x = -1$: $C = 1 - 3 + 5 = 3$.

> **Investigate further:** This is expressing $x^2 + 3x + 5$ in the *shifted basis* $\{(x + 1)^2, (x + 1), 1\}$ centred at $x = -1$. It is equivalent to the Taylor expansion about $x = -1$: $f(-1) = 3$, $f'(-1) = 1$, $f''(-1)/2! = 1$. Compare the coefficients with $A = f''(-1)/2$, $B = f'(-1)$, $C = f(-1)$.

**2.** The cubic $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0$, $p(-1) = 4$, $p(2) = 15$. Find $a, b, c$.

**Answer:** $a = \frac{11}{3},\ b = -3,\ c = -\frac{5}{3}$

**Method:** From $p(1) = 0$: $a + b + c = -1$. From $p(-1) = 4$: $a - b + c = 5$. Subtracting gives $b = -3$, then $a + c = 2$. Now $p(2) = 15$ gives $8 + 4a + 2b + c = 15 \Rightarrow 4a + c = 13$. Subtract $a + c = 2$: $3a = 11$, so $a = \frac{11}{3}$ and $c = -\frac{5}{3}$. So the coefficients are rational, not integers.

> **Investigate further:** Comparing coefficients (equating the polynomial at test points) is effectively solving a linear system $V\mathbf{c} = \mathbf{p}$ where $V$ is a *Vandermonde matrix*. The Vandermonde determinant $\prod_{i>j}(x_i - x_j)$ is nonzero as long as the evaluation points are distinct, guaranteeing a unique solution.

**3.** Simplify: $\dfrac{x^3 - 8}{x^2 + 2x + 4}$.

**Answer:** $x - 2$

**Method:** Difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. Cancel $(x^2 + 2x + 4)$.

> **Investigate further:** The factor $x^2 + 2x + 4$ has discriminant $4 - 16 = -12 < 0$, so it is irreducible over $\mathbb{R}$. This means $x = 2$ is the only real root of $x^3 = 8$. Generalise: $\frac{x^3-a^3}{x^2+ax+a^2} = x - a$ for any $a$.

**4.** State the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ and verify at $a = b = c = 1$.

**Answer:** At $a = b = c = 1$: LHS $= 3 - 3 = 0$; RHS $= 3(3 - 3) = 0$. ✓

**Method:** The factor $(a + b + c)$ on the RHS makes it obvious that if $a + b + c = 0$, the expression vanishes, so $a^3 + b^3 + c^3 = 3abc$. The identity is proved by expanding the RHS.

> **Investigate further:** The second factor $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2] \geq 0$.

> So the identity shows $a^3+b^3+c^3-3abc \geq 0$ when $a+b+c \geq 0$. Can you conclude $a^3+b^3+c^3 \geq 3abc$ for non-negative $a, b, c$? (Yes, since then $a+b+c \geq 0$.)

**5.** Given $a+b+c=0$, simplify $\dfrac{a^3+b^3+c^3}{abc}$.

**Answer:** 3

**Method:** Since $a+b+c=0$, by the identity above: $a^3+b^3+c^3=3abc$. So the ratio is 3.

> **Investigate further:** This ratio being a constant (3) regardless of the specific values of $a, b, c$ (subject to $a+b+c=0$) is a non-obvious result. Try $a=1, b=2, c=-3$: $1+8-27=3(1)(2)(-3)=-18$. $\frac{-18}{-6}=3$. ✓Now compute $\frac{a^4+b^4+c^4}{(a^2+b^2+c^2)^2}$ when $a+b+c=0$. (This was a WS#8 question; the answer is $\frac{1}{2}$.)

**6.** Verify $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n=1,2,3$; hence find $\sum_{k=1}^{10}(2k-1)^2$.

**Answer:** $\sum_{k=1}^{10}(2k-1)^2 = 1330$

**Method:** $\sum(2k-1)^2 = \sum(4k^2-4k+1) = 4\cdot\frac{10\cdot 11\cdot 21}{6} - 4\cdot\frac{10\cdot 11}{2} + 10 = 1540-220+10 = 1330$.

> **Investigate further:** The sum of squares of the first $n$ odd numbers is $\frac{n(2n-1)(2n+1)}{3}$. Verify for $n=3$: $1+9+25=35=\frac{3\cdot 5\cdot 7}{3}=35$. ✓Derive this formula from $\sum_{k=1}^{n}(2k-1)^2 = 4\sum k^2 - 4\sum k + n$.

**7.** Simplify: $\dfrac{\frac{1}{a+b}-\frac{1}{a-b}}{\frac{1}{a+b}+\frac{1}{a-b}}$.

**Answer:** $\dfrac{-b}{a}$

**Method:** Numerator: $\frac{(a-b)-(a+b)}{(a+b)(a-b)} = \frac{-2b}{a^2-b^2}$. Denominator: $\frac{2a}{a^2-b^2}$. Ratio: $\frac{-2b}{2a} = -\frac{b}{a}$.

> **Investigate further:** This is a *compound fraction* — a fraction whose numerator or denominator is itself a fraction. The strategy is always: simplify numerator and denominator separately before dividing. Now simplify $\dfrac{\frac{1}{x+h}-\frac{1}{x}}{h}$ (this is a difference quotient for $f(x)=\frac{1}{x}$).

**8.** Make $b$ the subject of $\dfrac{a-b}{a+b} = \dfrac{c-d}{c+d}$.

**Answer:** $b = \dfrac{ad}{c}$

**Method:** Cross-multiply: $(a-b)(c+d)=(c-d)(a+b)$. Expand: $ac+ad-bc-bd = ac+bc-ad-bd$. Simplify: $2ad=2bc \Rightarrow ad=bc \Rightarrow b=\frac{ad}{c}$.

> **Investigate further:** The condition $\frac{a-b}{a+b}=\frac{c-d}{c+d}$ is equivalent to $\frac{a}{b}=\frac{c}{d}$, i.e. the ratios $a:b$ and $c:d$ are equal. This is the *componendo-dividendo* rule: $\frac{p}{q}=\frac{r}{s} \Leftrightarrow \frac{p-q}{p+q}=\frac{r-s}{r+s}$. Prove it and use it to solve $\frac{x+3}{x-3}=\frac{7}{3}$ instantly.

**9.** Given $\alpha, \beta, \gamma$ are roots of $x^3-7x+6=0$, find $\alpha^2+\beta^2+\gamma^2$ and $\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2$.

**Answer:** $\alpha^2 + \beta^2 + \gamma^2 = 14; \quad \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = 49$

**Method:** Vieta: $\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = -7$, $\alpha\beta\gamma = -6$. $\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 0 - 2(-7) = 14$. $(\alpha\beta + \beta\gamma + \gamma\alpha)^2 = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma) = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 0$. So $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (-7)^2 = 49$.

> **Investigate further:** Notice that the sum of the roots is zero ($p = 0$ for the $x^2$ coefficient). This means the sum of squares is simply $-2q$. Use this shortcut to instantly find the sum of squares of the roots of $x^3 - 15x + 4 = 0$.

**10.** Solve: $\dfrac{2}{x^2 - 1} - \dfrac{1}{x - 1} = \dfrac{3}{x + 1}$.

**Answer:** No solution (the only candidate $x = 1$ is excluded from the domain).

**Method:** Multiply by $(x-1)(x+1)$: $2(x+1) - (x^2-1)/(x-1) = 3(x-1)$? No, $(x^2 - 1)$ is the common denominator. So $2 - (x + 1) = 3(x - 1) \Rightarrow 1 - x = 3x - 3 \Rightarrow 4x = 4 \Rightarrow x = 1$. But $x = 1$ makes the original denominators zero, so it is extraneous.

> **Investigate further:** This is the classic case where clearing denominators produces an extraneous solution. The lesson: *always check all solutions in the original equation.* The expression $\frac{2}{x^2-1} - \frac{1}{x-1}$ can be simplified to $\frac{-1}{x+1}$ for $x \neq 1$, making the equation $\frac{-1}{x+1} = \frac{3}{x+1}$, which has no solution. Spot this simplification first to avoid the extraneous solution trap.

## Section C Substitution & Structure

**1.** Solve: $\sqrt{x} - \dfrac{6}{\sqrt{x}} = 1$.

**Answer:** $x = 9$

**Method:** Let $t = \sqrt{x} > 0$: $t - \frac{6}{t} = 1 \Rightarrow t^2 - t - 6 = 0 \Rightarrow (t - 3)(t + 2) = 0$. Since $t > 0$: $t = 3$, so $x = 9$.

> **Investigate further:** The substitution $t = \sqrt{x}$ converts $f(\sqrt{x}) = 0$ to a polynomial equation in $t$. It must be combined with the condition $t \geq 0$ to discard negative roots. Try $\sqrt{x} + \frac{2}{\sqrt{x}} = 3$ and $\sqrt{x} - \frac{1}{\sqrt{x}} = \frac{3}{\sqrt{2}}$.

**2.** Find the minimum of $x^2 - 6x + y^2 + 4y + 20$.

**Answer:** Minimum is 7, at $(x, y) = (3, -2)$.

**Method:** $(x^2 - 6x + 9) + (y^2 + 4y + 4) + 20 - 9 - 4 = (x - 3)^2 + (y + 2)^2 + 7 \geq 7$. Equality at $x = 3$, $y = -2$.

> **Investigate further:** Geometrically, $(x-3)^2 + (y+2)^2 = r^2$ is a circle of radius $r$ centred at $(3, -2)$. The minimum of $(x-3)^2 + (y+2)^2$ is 0 at the centre. So the minimum of the expression is 7 (achieved at the centre). Now: what is the minimum of $x^2 - 6x + y^2 + 4y + z^2 - 2z + 20$?

**3.** Show $a^4 + b^4 \geq \frac{1}{2}(a + b)^4$ is false; prove the correct inequality $a^4 + b^4 \geq \frac{1}{2}(a^2 + b^2)^2$.

**Answer:** Counterexample: $a = b = 1$: $2 \geq \frac{1}{2}(16) = 8$? False. Correct inequality: $(a^2 - b^2)^2 \geq 0 \Rightarrow a^4 + b^4 \geq 2a^2b^2 \Rightarrow 2(a^4 + b^4) \geq (a^2 + b^2)^2$.

**Method:** $\frac{1}{2}(a+b)^4 = \frac{1}{2}(a^4+4a^3b+6a^2b^2+4ab^3+b^4)$. At $a=b=1$: $\frac{1}{2}(16) = 8 > 2 = a^4+b^4$. So the claimed inequality is false. Correct: $(a^2-b^2)^2 \geq 0$ gives $a^4 - 2a^2b^2 + b^4 \geq 0$, so $a^4+b^4 \geq 2a^2b^2$, hence $2(a^4+b^4) \geq a^4+2a^2b^2+b^4 = (a^2+b^2)^2$.

> **Investigate further:** The correct inequality $2(a^4+b^4) \geq (a^2+b^2)^2$ is a special case of the *power mean inequality*: $M_4 \geq M_2$, i.e. $\left(\frac{a^4+b^4}{2}\right)^{1/4} \geq \left(\frac{a^2+b^2}{2}\right)^{1/2}$. Verify this for $a=3, b=1$, then for $a=1, b=0$.

**4.** Solve $x^4 - 2x^3 - 3x^2 + 4x + 4 = 0$ given that it has a repeated real root. **Method:** Try $x=2$: $16-16-12+8+4=0$. ✓So $(x-2)^2$ divides the polynomial. Divide: $x^4-2x^3-3x^2+4x+4 = (x-2)^2(x^2+2x+1)$. So $(x-2)^2(x+1)^2 = 0$: $x=2$ or $x=-1$ (each a double root).

**Answer:** $x = 2$ (double root) and $x = -1$ (double root)

> **Investigate further:** A polynomial $p(x)$ has a repeated root $r$ iff both $p(r)=0$ and $p'(r)=0$ (the root is shared with the derivative). Check: $p'(x) = 4x^3-6x^2-6x+4$. $p'(2) = 32-24-12+4 = 0$ ✓. $p'(-1) = -4-6+6+4 = 0$ ✓. This derivative test is the standard method for finding repeated roots.

**5.** Given $a, b, c > 0$ with $a+b+c=1$, prove $ab+bc+ca \leq \frac{1}{3}$.

**Answer:** Proved below.

**Method:** $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca) = 1$. Also $a^2+b^2+c^2 \geq \frac{(a+b+c)^2}{3} = \frac{1}{3}$ (by C–S or AM inequality). So $2(ab+bc+ca) = 1-(a^2+b^2+c^2) \leq 1-\frac{1}{3} = \frac{2}{3}$, giving $ab+bc+ca \leq \frac{1}{3}$. $\square$ Equality iff $a=b=c=\frac{1}{3}$.

> **Investigate further:** Combined with the earlier result $ab+bc+ca \geq 0$, we have $0 \leq ab+bc+ca \leq \frac{1}{3}$ whenever $a,b,c>0$, $a+b+c=1$. The lower bound 0 is approached (but not achieved for positive reals) as one variable dominates. The upper bound $\frac{1}{3}$ is achieved at $a=b=c$.

**6.** If $x+y=1$, find the maximum of $x^2y+xy^2$.

**Answer:** $\dfrac{1}{4}$

**Method:** $x^2y+xy^2 = xy(x+y) = xy$. By AM–GM: $xy \leq \left(\frac{x+y}{2}\right)^2 = \frac{1}{4}$. Equality at $x=y=\frac{1}{2}$.

> **Investigate further:** The expression $x^2y+xy^2 = xy(x+y)$ factors instantly — this is the key recognition. Without seeing the factor, one might substitute $y=1-x$ and differentiate (using calculus), but the algebraic route is much faster. Always factor symmetric-looking expressions before substituting.

**7.** Solve over $\mathbb{R}$: $\sqrt{3x+1} - \sqrt{x+4} = 1$.

**Answer:** $x = 5$ only

**Method:** Isolate $\sqrt{3x+1} = 1+\sqrt{x+4}$ and square: $3x+1 = x+5+2\sqrt{x+4} \Rightarrow x-2 = \sqrt{x+4}$. Hence $x \geq 2$. Square again: $(x-2)^2 = x+4 \Rightarrow x^2-5x=0 \Rightarrow x=0$ or $x=5$. Domain $x \geq 2$ leaves $x=5$. Check: $\sqrt{16}-\sqrt{9}=1$.

> **Investigate further:** The squaring step introduced $x=0$ as an extraneous solution. When we required $x-2=\sqrt{x+4} \geq 0$, i.e. $x \geq 2$, this eliminated $x=0$ before even checking. Spotting the

> domain restriction *before* squaring saves a check step. Practise identifying domain restrictions at each squaring step.

**8.** For what values of $b, c$ do $x^2 + bx + c = 0$ and $x^2 + cx + b = 0$ have the same roots?

> **Answer:** $b = c$

**Method:** For both quadratics to have the exact same roots, all corresponding coefficients must be equal (since both are monic). Therefore $b = c$.

> **Investigate further:** The two equations have the *same* roots only when $b = c$. But they could each have *one common root*: if $r$ is a shared root, then $r^2 + br + c = 0$ and $r^2 + cr + b = 0$. Subtracting: $(b - c)r + (c - b) = 0 \Rightarrow (b - c)(r - 1) = 0$. So either $b = c$ or $r = 1$. If $r = 1$: $1 + b + c = 0$, so $b + c = -1$.

## Section D Challenge

(SMC / BMO1 difficulty)

**1. Schur's Inequality (degree 2):** Prove $\sum a^2(a-b)(a-c) \geq 0$ for $a, b, c \geq 0$.

> **Answer:** Proved below.

**Method:** WLOG $a \geq b \geq c \geq 0$. Group as $(a - b)[a^2(a - c) - b^2(b - c)] + c^2(c - a)(c - b)$. Since $c \geq 0$, $(c - a) \leq 0$ and $(c - b) \leq 0$, so the second term is $\geq 0$. For the first term, since $a \geq b \geq c \geq 0$, we have $a - c \geq b - c \geq 0$ and $a^2 \geq b^2$, so $a^2(a - c) \geq b^2(b - c)$. Thus the bracket is $\geq 0$, and $(a - b) \geq 0$. The sum is $\geq 0$.

> **Investigate further:** Schur's inequality for general $t \geq 0$ states $a^t(a - b)(a - c) + b^t(b - a)(b - c) + c^t(c - a)(c - b) \geq 0$. The $t = 1$ case gives: $a^3 + b^3 + c^3 + 3abc \geq (ab(a + b) + bc(b + c) + ca(c + a))$. This is a BMO1-level inequality. Look up the proof using $t = 1$ and try to verify it numerically for $a = 3, b = 2, c = 1$.

**2.** Find all polynomials with real coefficients satisfying $p(x^2) = p(x)^2$.

> **Answer:** $p(x) = 0$ and $p(x) = x^n$ for non-negative integers $n$.

**Method:** Suppose $p$ has degree $d$. Then $p(x^2)$ has degree $2d$ and $p(x)^2$ has degree $2d$: consistent. If $r$ is a root of $p$, then $p(r^2) = p(r)^2 = 0$, so $r^2$ is also a root. The sequence $r, r^2, r^4, \ldots$ must terminate (finitely many roots), so $r = 0$ or $r = 1$ or $r = -1$. If $r = -1$: $r^2 = 1$ is a root, and $1^2 = 1$, so $p(1) = p(1)^2 \Rightarrow p(1) = 0$ or 1. Careful analysis gives only monomials $p(x) = x^n$ and the zero polynomial.

> **Investigate further:** Functional equations for polynomials are a rich topic. Try $p(x + 1) - p(x) = nx^{n-1}$ (the discrete derivative). The polynomial that satisfies this has leading term $x^n$ (related to Bernoulli polynomials). Also explore: find all polynomials satisfying $p(x^2 - 1) = p(x)p(-x)$.

**3.** Find a closed form for $\dfrac{1^2}{1 \cdot 3} + \dfrac{2^2}{3 \cdot 5} + \cdots + \dfrac{n^2}{(2n - 1)(2n + 1)}$.

> **Answer:** $\dfrac{n(n + 1)}{2(2n + 1)}$

**Method:** $\frac{k^2}{(2k-1)(2k+1)} = \frac{1}{4} \cdot \frac{4k^2}{(2k-1)(2k+1)} = \frac{1}{4} \cdot \frac{(2k-1)(2k+1)+1}{(2k-1)(2k+1)} = \frac{1}{4}\left(1 + \frac{1}{(2k-1)(2k+1)}\right)$. Sum $= \frac{n}{4} + \frac{1}{4}\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)} = \frac{n}{4} + \frac{1}{4} \cdot \frac{n}{2n+1} = \frac{n}{4} \cdot \frac{2n+2}{2n+1} = \frac{n(n+1)}{2(2n+1)}$.

> **Investigate further:** Verify for $n = 1$: $\frac{1}{3} = \frac{1\cdot 2}{2\cdot 3} = \frac{1}{3}$. ✓For $n = 2$: $\frac{1}{3} + \frac{4}{15} = \frac{5+4}{15} = \frac{9}{15} = \frac{3}{5} = \frac{2\cdot 3}{2\cdot 5}$. ✓Now find $\sum_{k=1}^{n} \frac{k^2}{(k-1)(k+1)}$ for $k \geq 2$ using the same decomposition technique.

**4.** A polynomial $p(x)$ of degree 3 satisfies $p(n) = \frac{1}{n}$ for $n = 1, 2, 3, 4$. Find $p(5)$.

**Answer:** $p(5) = 0$

**Method:** Consider $q(x) = xp(x) - 1$. Then $q(n) = n \cdot \frac{1}{n} - 1 = 0$ for $n = 1, 2, 3, 4$, so $q$ is a quartic with roots $1, 2, 3, 4$: $q(x) = c(x-1)(x-2)(x-3)(x-4)$. At $x = 0$: $q(0) = -1 = 24c$, so $c = -\frac{1}{24}$. Hence $q(5) = -\frac{1}{24}(4)(3)(2)(1) = -1$. But $q(5) = 5p(5) - 1$, so $5p(5) - 1 = -1$, giving $p(5) = 0$.

> **Investigate further:** So $p(5) = 0$ means $x = 5$ is a root of $p$. This is a striking result! Explore: compute $p(0)$. From $q(0) = 0 \cdot p(0) - 1 = -1$ and $q(0) = c \cdot (-1)(-2)(-3)(-4) = 24c = -1$, so $c = -\frac{1}{24}$. Then $p(x) = \frac{q(x)+1}{x} = \frac{-\frac{1}{24}(x-1)(x-2)(x-3)(x-4)+1}{x}$ for $x \neq 0$.

**5.** Find all pairs of positive integers $(m, n)$ with $m > n$ such that $m^2 - n^2 = 105$.

**Answer:** $(m, n) \in \{(53, 52),\ (19, 16),\ (13, 8),\ (11, 4)\}$

**Method:** $(m-n)(m+n) = 105 = 3 \times 5 \times 7$. Set $m - n = a$, $m + n = b$: $ab = 105$, $b > a > 0$, same parity (both odd since 105 is odd and $a, b$ have same parity). Factor pairs of 105 with $a < b$, both odd: $(1, 105), (3, 35), (5, 21), (7, 15)$. Each gives $m = \frac{a+b}{2}$, $n = \frac{b-a}{2}$: $(1, 105) \to (53, 52)$; $(3, 35) \to (19, 16)$; $(5, 21) \to (13, 8)$; $(7, 15) \to (11, 4)$.

> **Investigate further:** The key observation: since 105 is odd, $m - n$ and $m + n$ must both be odd (both even would give $4 \mid (m^2 - n^2)$, but $4 \nmid 105$). This parity constraint halves the number of cases. Generalise: for which $N$ does $m^2 - n^2 = N$ have no positive integer solutions? (When $N \equiv 2 \pmod 4$ or $N = 1$ or $N = 4$.)

## Top 5 Patterns — Worksheet #6

1. **Shifting basis: $p(x)$ in powers of $(x - a)$.** Evaluate $p(a)$, $p'(a)$, $p''(a)$ etc. to find coefficients without expanding. Equivalent to Taylor expansion at $x = a$.

2. **Substitution $t = \sqrt{x}$ for surd equations.** Always combine with the constraint $t \geq 0$ to discard extraneous roots before they enter the algebra.

3. **The identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(\cdots)$.** The zero of the first factor $(a + b + c = 0)$ is the most useful case. Memorise both the identity and the special case.

4. **Polynomial masking: $q(x) = xp(x) - 1$ for rational interpolation.** When $p(n) = 1/n$, multiply through to clear denominators, find the masked polynomial's roots, determine its leading coefficient from $q(0) = -1$.

5. **Factoring $m^2 - n^2 = (m - n)(m + n)$ and using parity.** The parity of $m \pm n$ is determined by the parity of $m^2 - n^2$. Use this to eliminate half the factor pairs immediately.

## Common Traps — Worksheet #6

1. **Extraneous solutions after clearing denominators.** In B10, $x = 1$ satisfies the cleared equation but not the original. Always substitute back.

2. **Not checking both cases when squaring radical equations.** In C7, squaring $x - 2 = \sqrt{x + 4}$

requires $x \geq 2$: this eliminates $x = 0$ before checking.

3. **Confusing "same roots" with "same polynomial."** In C8, two quadratics can share all roots only if all coefficients are equal (for monic quadratics). Carefully set up the Vieta conditions for both.

4. **Applying $a^4 + b^4 \geq \frac{1}{2}(a+b)^4$.** This is false. The correct form is $2(a^4 + b^4) \geq (a^2 + b^2)^2$, which is weaker. Test inequalities with $a = b = 1$ before claiming them.

5. **Forgetting that $p(x^2) = p(x)^2$ constrains roots severely.** The functional equation forces the root set to be closed under squaring, leaving only $\{0\}, \{1\}, \{0, 1\}$, or empty. Don't guess a general polynomial form.

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*"In mathematics the art of proposing a question must be held of higher value than solving it." — Georg Cantor*

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Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths